Growth and Performance Analysis

Acqusition-Drive Growth: A Marketplace Yet to Build a Retention Engine

Brazilian E-Commerce Public Dataset by Olist
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Acquisiton-Driven Growth: A Marketplace Yet to Build a Retention Engine

Executive Summary

This analysis examines wheather the growth of a transactional marketplace is driven by a stable and expanding retention base, or whether it remains structurally dependent on continuous new customer acquisition. The findings reveal a business that is growing in volume but has yet to develop the retention foundations neccessary for sustainable, long-term growth.

Objective

This analysis aim to determine whether marketplace growth is ther result of stable or improving customer retention, high-quality acquisition that converts into repeat behavior, or variables that are inherently difficult to sustain over time. To answerthis, the analysis is structured around five business questions covering revenue contribution, cohort decay, revenue versus customer retention, behavioral expansion, and structrural churn leakage.

Key Findings

- Revenue growth is almost entirely driven by new customer acquisition, with new customers contributing 97% of total revenue throughout the observed period, leaving the existing customer base with a marginal and stagnant share.
- Retention collapses immediately after the first purchase, with more than 99% of customers not returning in the following month. This pattern is consistent across all cohorts and shows no signs of improvement over time. It is worth noting that this marketplace is dominated by durable goods with medium to long usage cycles, meaning a portion of this pattern may reflect natural buying behavior rather than pure retention failure, customers who purchased furniture, construction tools, or electronics may simply not need to repurchase within the two-year observation window.
- Nearly half of all existing customer transactions, at 49.94%, occur outside the normal purchase cycle, falling within the at-risk and churned zones. While this indicates that retention is not happening proactively, the durable goods context suggests that not all customers in these zones have genuinely disengaged, some may be operating within a natural buying cycle for their product category. What remains clear is that the business currently lacks the infrastructure to distinguish between the two.

- Existing customers spend less than new ones, with retained and resurrected customers generating an AOV that is 8 to 10% lower than new customers. Rather than confirming outright monetization failure, this gap may partly reflect a natural shift in purchase composition. New customers tend to make larger initial purchases such as furniture or electronics, while returning customers are more likely to buy complementary products at lower price points. Regardless of the cause, no upward movement in existing customer AOV has been observed over time.
- Revenue retention marginally outpaces customer retention, suggesting that returning customers spend more per visit than in the prior period. However, given the dominance of durable goods, this may partly reflect category-switching behavior rather than genuine loyalty-driven expansion. The base remains too small for this signal to generate meaningful business impact at scale.

Business Implication

The marketplace is currently operating on an acquisition-led growth model, where revenue is sustained by continuous inflow of new customers rather than a compounding base of loyal ones. This creates a structurally fragile business. Any slowdown in acquisition directly translate to a revenue decline, with no retained base to absorb the impact.

The dominance of durable goods across the product mix adds an important layer of complexity to this picture. In a marketplace where the majority of products have medium to long usage cycles, the conventioanl retention playbook that built around repurchase within a short window may not be directly applicable. Customers who have purchased furniture, construction tools, or electronics may not need to buy again for months or years, which means standard re-engagement approaches risk being poorly timed and therefore ineffective. At the same time, this characteristic also represents an untapped opportunity, if the business can successfully drive cross-category purchaseing among existing customers, it can generate incremental revenue without waiting for the natural repurchase cycle to complete.

Without a deliberate shift toward a more nuanced retention strategy, one that accounts for both product lifecycle and customer engagement the business will face growing pressure on unit economics and limited expansion of customer lifetime value.

Recommendations

- Prioritize post-purchase engagement by implementing onboarding sequences, personalized follow-ups, and loyalty incentives to convert first-time buyers into repeat customers. This directly addresses the acquisition dependency identified in the analysis and is the most immediate lever available to the growth team.
- Identify the critical window between the first and second purchase and deploy targeted re-engagement campaigns, including time-sensitive offers, category recommendations based on first purchase behavior, and satisfaction follow-ups. Acting within this window before customers enter the at-risk zone is significantly more cost-effective than reactivating customers who have already churned.
- Focus on growing the retained customer base before shifting resources toward spending expansion programs. Until retention volume reaches a meaningful scale, upsell and cross-sell initiatives will have limited measurable impact on overall revenue performance.

Data & Methodology

This analysis uses the Brazilian E-Commerce Public Dataset by Olist, publicly available on Kaggle. The dataset is a relational database consisting of multiple tables covering orders, customers, payments, and products across a two-year observation period.

The analysis combines revenue evaluation, customer cohort analysis, and customer spend analysis to assess marketplace performance and identify the underlying growth engine. Customer segments are defined as new customer, retained customer, and ressurected customer. All metrics and findings are derived from actual transaction-level data.

Background

The e-commerce industry has long recognized customer retention as one the most critical drivers of sustainable growth. While acquiring new customers is essential in the early stages of a marketplace, long-term profitability and revenue stability depend heavily on the ability to convert first-time buyers into loyal, repeat customers. Businesses that fail to build a strong retention engine inevitably find themselves trapped in a costly cycle of continuous acquisition, where growth is possible but increasingly inefficient over time.

This project was initiated to examine whether growth in a transactional marketplace is truly sustainable, or whether it remains disproportionately dependent on new customer acquisition. By analyzing customer behavior patterns across multiple dimensions, this project aims to surface structural weaknesses in the retention lifecycle and provide a foundation for more informed, data-driven decisions around customer engagement strategy.

This project is designed to answer one overaching question: is the growth of this marketplace sustainable and retention-driven, or is it still acqisition-dependent?
To answer this, the analysis is structured around five business questions:

1. How large is the revenue contribution gap between new and existing customers?
2. How quickly do cohorts decay after the first purchase?
3. Does revenue retention outperform customer retention?
4. Is there evidence of behavioral expansion among existing customers?
5. How significant is structural churn leakage based on purchase inactivity?

The analysis covers customer transaction data from October 2016 to August 2018. The analysis focuses on three customer segments: new customers, retained customers, and resurrected customers, with all metrics and findings derived from actual transaction-level data.

Objective

The primary objective of this analysis is to determine whether the growth of this marketplace is genuinely sustainable or primarily dependent on continuous customer acquisition. specially, this analysis aims to:

- Identify the underlying growth engine of the marketplace by evaluating the revenue contribution first tansaction split between new and existing customers across the observed period.
- Analyze customer purchase behavior to understand whether customers return after their first transaction and, if so, how quickly retention deteriorates over time.
- Assess the degree of behavioral expansion among existing customers by examining whether returning customers demonstrate increased spending over time.
- Quantify the structural churn leakage within the existing customer base to identify where engagement is lost and where intervention is most likely to be effective.
- Provide data-driven recommendations that give the growth team a clear, actionable foundation for evaluating marketplace performance and prioritizing retention initiatives.

Methodology

The analysis was conducted in a structured manner, combining data extraction, exploratory analysis, feature engineering, and performance evaluation to assess marketplace growth from both a revenue and customer behavior persprective.

1. **Data Extraction**

The first step involved extracting relevant data form a relational database using Microsoft SQL Server. Queries were designed to retrieve transaction-level records neccessary for customer segmentation, retention analysis, and revenue evaluation.

2. **Exploratory Data Analysis**

Following data extraction, an exploratory analysis was conducted to understand the structure and distribution of the data. This include summary statistics and distribution analysis for key variables, particularly interpurchase day, to identify patterns that would inform feature definition and threshold setting.

3. **Feature Definition**

Key analytical metrics and segment were defined to provide a consistent foundation for the analysis:

- **Customer Segment:** Three segments were defined based on purchase history. New customers are thos making their first-ever purchase on the platform. Retained cusotmers are those who return within a normal purchase cycle after their first transaction. Ressurected customers are those who had been inactive for an extended period before returning to make another purchase.
- **Interpurchase Day Threshold:** Thresholds were derived from the statistical distribution of the interpurchase day variable, specifically t**he median (29 days), 75th percentile (122 days), and 90th percentile (240 days).** These thresholds define three customer activity zones: active, at-risk, and churned, providing a data-driven basis for classifying cusotmer engagement levels.
- **Customer Retention Rate (CRR) and Revenue Retention Rate (RRR):** Both metrics were calculated using the same base of existing customers, ensuring a consistent and comparable measurement. When RRR exceeds CRR, it indicates that returning customers are generating higher revenue per period than before, signaling a degree of spending expansion within the retained base. Given that Olist operates as a transactional marketplace rather than a subscription-based business, both metrics represent an adaptation of the conventional CRR and RRR framework to fit a non-recurring purchase context.

4. **Data Characteristics & Limitation**

An additional contextual factor worth noting is the nature of the products sold on this marketplace. A review of product categories with average interpurchase days exceeding 100 days reveals that the majority consist of products with medium to long usage cycles, including furniture, construction tools, musical instruments, and computers. More broadly, the overall product mix on this marketplace is dominated by durable goods rather than fast-moving consumer products. This has meaningful implications for how the retention findings in this analysis should be interpreted. In markets dominated by durable goods, a customer not returning within a one to two year observation window does not necessarily indicate disengagement, it may simply reflect that the product purchased has not yet reached the end of its useful life. This limitation applies most directly to the findings in customer cohort decay analysis and churn leakage analysis, and should be kept in mind when reading those sections.

5. **Analytical Approach**

The analysis was structured around four dimensions. Revenue contribution was evaluated by comparing the share of revenue generated by new customers against existing customers across the observed period. Customer behavior was assessed through cohort analysis to measure retention decay over time. Customer value was examined through Average Order Value comparison across segments. Structural churn leakage was quantified by classifying existing customer transactions into the three activity zones defined above.

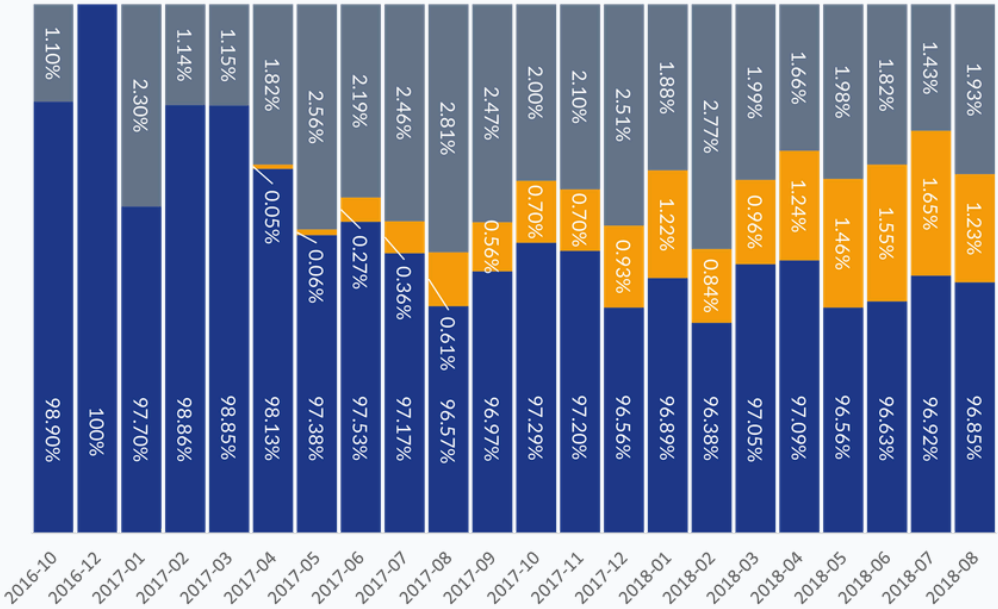
6. **Tools**

Microsoft SQL Server was used to extract and query data from the relational database. Python was used for exploratory data analysis, including distribution analysis and summary statistics. Microsoft Excel was used to conduct the core analysis and build supporting visualizations. Power BI was used to develop the interactive dashboard for performance monitoring.

New Customers Generate 97% of Revenue While the Existing Base Contributes Almost Nothing

Revenue growth is overwhelmingly acquisiton-driven, with minimal contribution from existing customers.

New Customer	Retained	Resurrected	Acquisition Dependency
97.4%	**1.9%**	**0.7%**	**97.4%**
of total revenue throughout the period	Stagnant, with no significant growth	Grew from 0% to 1.6% (Jul-18)	Revenue is dependent on new buyers



- New customers dominate revenue at 96-99% throughout the entire observed period, a pattern that has not meaningfully shifted, signaling that repeat purchaseing behavior has yet to take hold at scale.
- Retained and resurrected customers combined contribute less than 4% of total revenue, with no clear upward trajectory. Indicating that current retention and reactivation efforts are not moving the needle.
- Notably, resurrected customers show a slight growth from ~0.2% (H1 2017) to ~1.6% (mid-2018), while retained customers remain flat at ~2%, suggesting that what little existing customer growth exist is driven by dormant users returning, not by a loyal repeat buyers.
- A business generating 97% of revenue from new customers is structurally fragile, any slowdown in acquisition directly translates to a revenue decline, with no retention base to absorb the impact.
- Customer Acquisition Cost will continue to rise as the easiest-to-reach customers are acquired first. Without a growing retained base to offset this, revenue growth becomes increasingly expensive to sustain.

Retention Collapses After the First Purchase With No Cohort Showing Signs of Recovery

Customer cohorts decay almost entirely after the first puchase, indicating failure in second-purchase conversion.

AVG Retention M1
0.48%
Average across all cohorts

Decay M0 → M1
> 99%
Almost all customers do not return

Cohort Terkuat M1
2017-04
0.66%, highest at M1

Month Cohort	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	19	20
2016-10	100%						0.38%			0.38%		0.38%		0.38%		0.38%		0.38%	0.75%	0.75%
2016-12	100%	100%																		
2017-01	100%	0.27%	0.27%	0.13%	0.40%	0.13%	0.40%	0.13%	0.13%		0.40%	0.13%	0.67%	0.40%	0.13%	0.13%	0.27%	0.40%	0.13%	
2017-02	100%	0.18%	0.30%	0.12%	0.42%	0.12%	0.24%	0.18%	0.12%	0.18%	0.12%	0.30%	0.12%	0.18%	0.12%	0.06%	0.06%	0.18%		
2017-03	100%	0.47%	0.35%	0.47%	0.35%	0.16%	0.16%	0.31%	0.31%	0.08%	0.35%	0.12%	0.20%	0.12%	0.16%	0.24%	0.08%	0.12%		
2017-04	100%	0.66%	0.26%	0.22%	0.26%	0.26%	0.39%	0.31%	0.31%	0.18%	0.26%	0.09%	0.04%	0.04%	0.09%	0.09%	0.18%			
2017-05	100%	0.48%	0.45%	0.28%	0.28%	0.34%	0.45%	0.14%	0.26%	0.31%	0.26%	0.34%	0.23%	0.03%	0.17%	0.20%				
2017-06	100%	0.52%	0.42%	0.42%	0.32%	0.39%	0.36%	0.23%	0.13%	0.19%	0.29%	0.39%	0.16%	0.19%	0.32%					
2017-07	100%	0.58%	0.34%	0.24%	0.34%	0.21%	0.31%	0.10%	0.21%	0.29%	0.21%	0.31%	0.13%	0.24%						
2017-08	100%	0.70%	0.39%	0.27%	0.34%	0.56%	0.29%	0.27%	0.15%	0.15%	0.24%	0.19%	0.12%							
2017-09	100%	0.69%	0.54%	0.37%	0.44%	0.32%	0.22%	0.25%	0.27%	0.17%	0.25%	0.07%								
2017-10	100%	0.75%	0.25%	0.09%	0.25%	0.21%	0.21%	0.39%	0.30%	0.18%	0.21%									
2017-11	100%	0.59%	0.38%	0.17%	0.17%	0.20%	0.13%	0.18%	0.13%	0.06%										
2017-12	100%	0.20%	0.30%	0.37%	0.26%	0.20%	0.17%	0.02%	0.19%											
2018-01	100%	0.33%	0.36%	0.29%	0.29%	0.17%	0.17%	0.23%												
2018-02	100%	0.34%	0.40%	0.30%	0.25%	0.22%	0.20%													
2018-03	100%	0.39%	0.34%	0.31%	0.12%	0.12%														
2018-04	100%	0.59%	0.30%	0.24%	0.15%															
2018-05	100%	0.53%	0.26%	0.18%																
2018-06	100%	0.42%	0.27%																	
2018-07	100%	0.52%																		
2018-08	100%																			

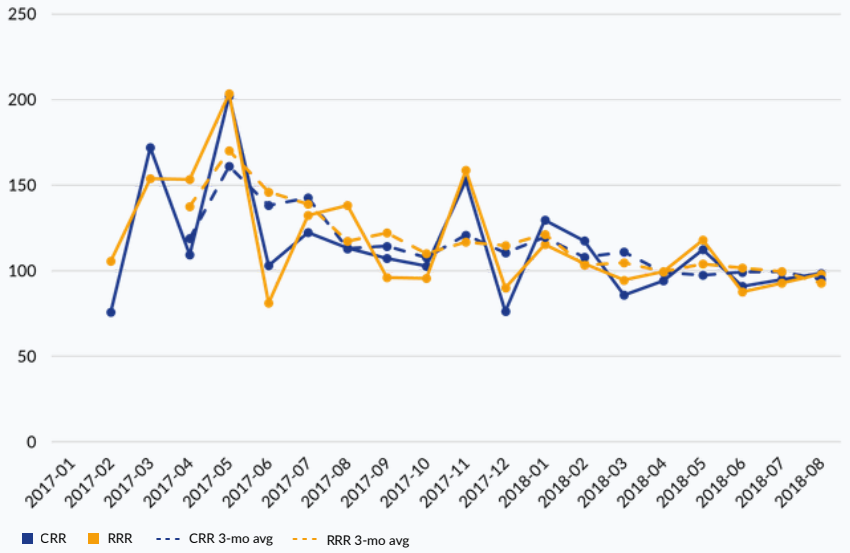
Note: 2016-12: M1 retention of 100% is not an indicator of loyalty, as this cohort consists of only one customer and is therefore not representative as a benchmark. Aug-2018 data is included with a note that the retention rate for this cohort cannot be calculated (observation period is incomplete). Index 18 does not appear as no customers returned during that period.

- Every cohort recorderd across the observed period experiences a retention collapse immediately after the first purchase, with more than 99% of customers not returning in the following month, making one-time purchase the dominant behavior across the entire marketplace.
- Retention levels remain consistently below 1% throughout the customer lifecycle, with no cohort demonstrating a meaningful recovery at any point after M1. At face value, this points to a severe reteention problem. However, given that the marketplace is predominatnly selling durable goods with medium to long usage cycles, a portion of this pattern may reflect natural buying behavior rather than pure disengagement. Customers who purchase furniture, construction tools, or electronics may simply not need to repurchase within the two-year observation window.
- That said, the absence of improvement across newer cohorts remains a meaningful signal. If the pattern were purely dirven by product nature, we would still expect to see some cohorts performing better than others as the customer base grows and diversifies. The consistency of near-zero retention across all cohorts suggest that structural factors beyond product type are also at play. Including the absence of post-purchase engagement, loyalty programs, or mechanisms to drive cross-category purchasing.
- The one-time purchase pattern limits the ability bo build a compounding revenue base regadless of whether it is driven by product nature or retention failure. Even if some customers are simply operating within a long buying cycle, the business has no mechanism to capture their next purchase when it does occur, nor to encourage them to explore other categories in the interim.
- Without the ability to distinguish between customers who have churned and thos who are simply in a long buying cycle, the business cannot efficiently allocate re-engagement resources. Blanket reactivation campaigns risk targeting customers who were never disengaged, while missing those who have genuibely moved to competitors.

Returning Customers Spend More Per Visit But the Base Is Too Small to Matter

Revenue retention slightly outperforms customer retention, but is driven by a small base of returning customers

AVG CRR (Feb-17 - Aug-18)	AVG RRR (Feb-17 - Aug-18)	Gap RRR VS CRR
113.08%	116.07%	+ 2.99 pp
Monthly average of existing customers	Monthly average of existing revenue	Revenue grows faster than customer count



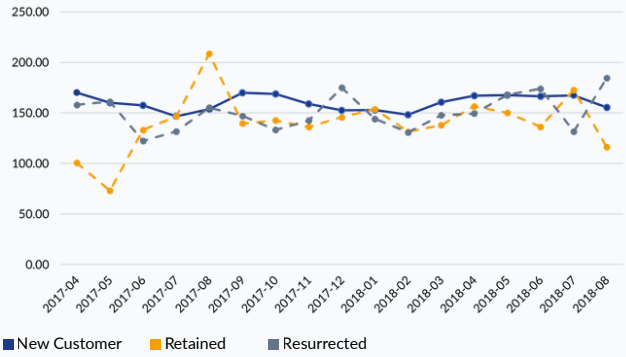
Note: CRR and RRR are calculated as the ratio of month N to month N-1 from existing customers (Retained + Resurrected). Values above 100% can occur due to spikes in Resurrected customers in certain months. Data from Oct-2016 and Jan-2017 are excluded. Oct-2016 is excluded due to unrepresentative volume (n=3), while Jan-2017 is excluded as it has no N-1 as a denominator. These metrics are an adaptation of CRR/RRR for a transactional e-commerce context, rather than a subscription-based one.

- Revenue Retention Rate (RRR) marginally outpaces Customer Retention Rate (CRR), averaging 116.81% and 113.78% respectively. This indicating that returning customers tend to transact at a higher value than in the prior period. However, given that this marketplace is dominated by durable goods, this pattern may partly reflect customers returning to purchase different or complementary categories rather than demonstrating genuine loyalty-driven spending expansion within the same category.
- Both metrics display significant volatility, swinging between 75% and over 200% across months. This is not reflection of erracting customer begavior, but a direct consequence of the extremely small returning customer base, where even a handful of additional or missing transactions produces outsized percentage movements. The dominance of durable goods further amplifies this volatility, as single high-value transactions in categories like furniture or electronics can disproportionately shift the monthly average.
- The positive gap between RRR and CRR, while encouraging, should be interpreted with caution. It signals that the few customers who do return are spending more, but whether this reflects genuine behavioral expansion or simply the natural variation in purchase size across product categories cannot be determined from the current data alone.
- A business where RRR consistently exceeds CRR has the foundation for a healthy expansion revenue engine. However, this signal only becomes actionable at scale. With a returning base this thin, the expansion effect is statistically fragile and easily distorted by one or two large transactions, particularly in a durable goods context where individual order values can vary significantly.
- Revenue durability remains highly sensitive to fluctuations in a very small customer segment. Any month-to-month variation in this group disproportionately impacts both CRR and RRR, making these metrics unreliable as standalone indicators of business health without also growing retention volume.

Existing Customers Spend Less Than New Ones With No Behavioral Expansion Detected

Existing customers show lower AOV, indicating failure to grow customer lifetime value

AOV New Customer	AOV Retained	AOV Resurrected
160.31	147.21	144.58
reference baseline | 93,359 orders	-8.2% vs new customer | 2,149 orders	-9.8% vs new customer | 971 orders



Segment	AOV	Orders	Gap vs New
New Customer	160.31	93,359	-
Retained	147.21	2,149	-13.10
Resurrected	144.58	971	-15.73

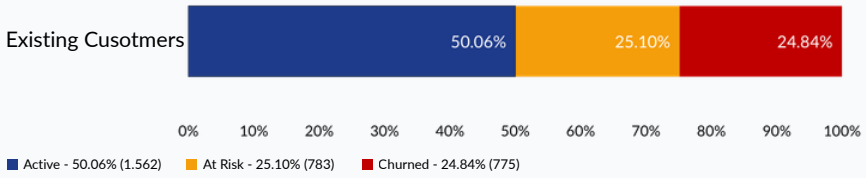
Note: The chart starts from Apr-2017 as there were no transactions from the Resurrected segment prior to that period. Oct-2016 and Dec-2016 are excluded from the chart due to unrepresentative transaction volume (n ≤ 3).

- Retained and resurrected customers consistently generate a lower Average Order Value than new customers throughout the entire observed period, with a gap of 8 to 10%, appearing to contradict the common expectation that familiarity with a platform leads to higher spending over time.
- Retained customers average an AOV of 147.21 and resurrected customers 144.58, compared to 160.31 for new customers. This gap remains stable across months with no signs of convergence, suggesting a persistent pattern rather than a temporary anomaly.
- The product context established earlier offers a partial explanation for this gap. New customers making their first purchase may tend to buy primary, higher-value items such as furniture or electronics, while returning customers are more likely to purchase complementary or secondary products at lower price points. This means the AOV gap may not be purely a failure of monetization, it may partly reflect a natural shift in purchase composition as customers move through their lifecycle on the platform.
- That said, the absence of any upward movement in existing customer AOV over time still indicates that the marketplace has not successfully created conditions for spending expansion whether through product discovery, personalized recommendations, or loyalty-driven incentives. Even accounting for the product mix effect, we would expect at least some cohorts to show AOV growth if engagement mechanisms were working effectively.
- The combination of low retention volume and below-average AOV leaves the revenue contribution of existing customers structurally limited. This compounds the acquisition dependency identified in the revenue contribution analysis, not only are there too few returning customers, but those who do return spend less, further constraining lifetime value regardless of whether this is driven by product nature or engagement failure.
- Without understanding whether the AOV gap is driven by product composition or genuine under-monetization, the business cannot design effective interventions. Treating it purely as a monetization failure risks misallocating resources toward upsell programs that may not be relevant for a durable goods customer base.

Nearly Half of Existing Customer Transactions Occur Outside the Normal Purchase Cycle

Nearly half of transaction occur after customer fall outside normal purchase behavior

Active Zone (≤ 29 Days)	At Risk Zone (30-122 Days)	Churn Leakage Total
50.06%	25.10%	49.94%
1.562 from 3.120 transactions	783 transactions	at risk + churned



Zone	Range	Threshold	Transactions	%
Active	1 - 29 Days	≤ Median	1,562	50.06%
At Risk	30 - 122 Days	75th pct	783	25.10%
Churned	> 122 Days	> 75 pct	775	24.84%
Total			3,120	100%

Note: Analysis is based on existing customers (Retained + Resurrected), covering a total of 3,120 transactions with recorded interpurchase days. The threshold is determined from the statistical distribution of interpurchase days: median = 29 days, 75th percentile = 122 days, 90th percentile = 241 days. New customers (interpurchase day = null) are excluded.

- Nearly half of all transaction from existing customers, 49.94% in total, occur after customer have already fallen outside their normal purchase cycle, split between the at risk zone at 25.10% and the churned zone at 24.84%, indicating that retention is not happening consistently, but through delayed reactivation.
- However, this finding must be read alongside the product context established in hte methodology. The interpurchase day thresholds used 29 days for active, 122 days for at-risk, and 241 days for churned were derived from the overall distribution across all categories without accounting the product type. Given that the marketplace is dominated by durable goods, a meaningfull portion of transaction classified as at-risk or churned may reflect customer operating within a natural buying cycle for their product category rather than true disengagement.
- Dispite this caveat, the structural pattern remains significant. Only 50.06% of existing customer transaction occur within the acrive zone, meaning that even among customers who eventually return, the majority do so after an extended period of inacrivity. Whether this is driven by product nature or retention failure, the business currently has no mechanism to differentiate between the two and therefore cannot act on either efficiently.
- Churn leakage of this magnitude, even when partially explained by product characteristcs, signals a gap in customer engagement infrastructure. The business is not proactively maintaining relationship with its existing base, leaving it unable to capture purchase intent when it does arise whether for repurchase or cross-category exploration.
- Without category-aware sementation, re-engagement resources are likely being spread to broadly. Customers in long buying cycle categories who are contacted too early may find the outreach irrelevant, reducing campaign effectiveness and potentially damaging the relationship. Meanwhile, customers in faster-moving categories who genuinely need a nudge may not be receiving timely attention.

Summary & Recommendations

Summary:
The analysis confirms that growth in this marketplace is almost entirely driven by new customer acquisition, while the existing customer base contributes very little to overall revenue. This pattern is not inherently problematic in the short term, as acquisition-led growth can sustain revenue momentum when customer inflow remains strong. However, if this dependency persist without meaningful improvement in retention, the business will face increasingly serious structural challanges as acquisition costs rise over time.

An imprtant contextual factor that shapes how these findings should be read is the nature of the products sold on this marketplace. The product mix is dominated by durable goods with medium to long usage cycle, including furniture, construction tools, electronics, and musical instruments. This means that a portion of the low retention and high churn leakage figures observed in this analysis may reflect natural buying behavior rather than pure retention failure. Customers who purchased durable goods may simply not need to repurchase within the two-years observation window. This does not invalidate the core findings, but it does add an important layer of nuance to how the numbers should be interpreted and acted upon.

What remains clear regardless of product context is that the business currently lacks the infrastructure to distinguish between customers who have genuinely churned and those who are simply in a long buying cycle. Without this capability, re-engagement efforts cannot be efficiently targeted, and the opportunity to drive cross-category purchasing among existing customers remains largely untapped.

Key insights:

1. Revenue growth is almost entirely driven by new customer acquisition, with new customers contributing 97% of total revenue throughout the observed period, leaving the existing customer base with a marginal and stagnant share.
2. Retention collapses immediately after the first purchase, with more than 99% of customers not returning in the following month. This pattern is consistent across all cohorts and shows no signs of improvement over time. It is worth noting that this marketplace is dominated by durable goods with medium to long usage cycles, meaning a portion of this pattern may reflect natural buying behavior rather than pure retention failure, customers who purchased furniture, construction tools, or electronics may simply not need to repurchase within the two-year observation window.
3. Nearly half of all existing customer transactions, at 49.94%, occur outside the normal purchase cycle, falling within the at-risk and churned zones. While this indicates that retention is not happening proactively, the durable goods context suggests that not all customers in these zones have genuinely disengaged, some may be operating within a natural buying cycle for their product category. What remains clear is that the business currently lacks the infrastructure to distinguish between the two.
4. Existing customers spend less than new ones, with retained and resurrected customers generating an AOV that is 8 to 10% lower than new customers. Rather than confirming outright monetization failure, this gap may partly reflect a natural shift in purchase composition. New customers tend to make larger initial purchases such as furniture or electronics, while returning customers are more likely to buy complementary products at lower price points. Regardless of the cause, no upward movement in existing customer AOV has been observed over time.
5. Revenue retention marginally outpaces customer retention, suggesting that returning customers spend more per visit than in the prior period. However, given the dominance of durable goods, this may partly reflect category-switching behavior rather than genuine loyalty-driven expansion. The base remains too small for this signal to generate meaningful business impact at scale.

Summary & Recommendations

Recommendations:

1. Prioritize post-purchase engagement by implementing onboarding sequences, personalized follow-ups, and loyalty incentives to convert first-time buyers into repeat cusotmers. This directly addresses the acquisition dependency identified in the analysis and is the most immediate lever available to the growth team.
2. Develop a category-aware engagement strategy that accounts for natural buying cycles. For cusotmers who purchased durable goods such as furniture, construction tools, or electronics, focus on cross-category discovery and value-added content rather than repurchase prompts. For customers in faster-moving categories, deploy proactive re-engagement within the established interpurchase day threshold windows. This distinction will improve campaign relevance, reduce wasted spend, and produce cleaner retention data over time.
3. Identify the critical window between the first and second purchase for fast-moving category buyers and deploy targeted re-engagement campaigns, including time-sensitive offers, category recommendations based on first purchase behavior, and satisfaction follow-ups. Acting within this window before customers enter the at-risk zone is significantly more cost-effective than reactivating customers who have already churned.
4. Focus on growing the retained customer base before shifting resources towards spending expansion programs. Until retention volume reaches a meaningful scale, upsell and cross-sell initiative will have limited measurable impact on revenue performance.